Function transformations

1 How should you translate the graph of $y = f\,(x)$ to get the graph of $y = f\,(x) + 4$?

2 How should you translate the graph of $y = g(x)$ to get the graph of $y = g(x + 6)$?

3 Describe the shift required to transform the graph of $y = a^x$ to get the graph of $y = a^{(x+5)}$.

4 For each of the following equations, determine what the new equation is when their graphs are moved as described:

a The graph of $y = 4x$ is moved up by 2 units.

b The graph of $y = x^3$ is moved to the right by 10 units.

c The graph of $y = 2^x$ is moved down by 9 units.

5 Suppose that the x-values of the x-intercepts of the graph of $y = f\,(x)$ are -5 and 6. Find the corresponding x-values of the x-intercepts of the graph of:

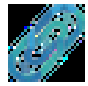
a $y = f\,(x + 4)$ b $y = f\,(x - 4)$ c $y = 3f\,(x)$ d $y = f\,(-x)$

6 State the vertical stretch factor of:

a $y = 4x^3$ b $y = -4x^4$ c $y = 3x^5$ d $y = -5x^2$

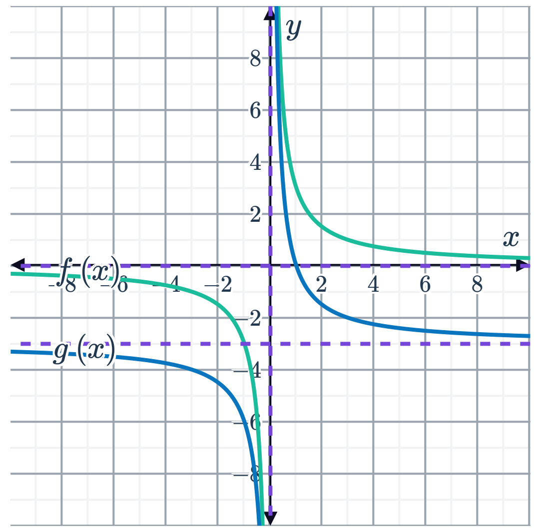
7 Consider the function $f\,(x) = x^2 - 5$. Using function notation, describe the transformation of f that will result in the function:

a $g\,(x) = x^2 - 6$ b $h\,(x) = 16x^2 - 5$ c $k\,(x) = 3x^2 - 15$

8 Consider the graphs of $f(x) = \frac{3}{x}$ and $g(x)$: 

a Write $g(x)$ in terms of $f(x)$.

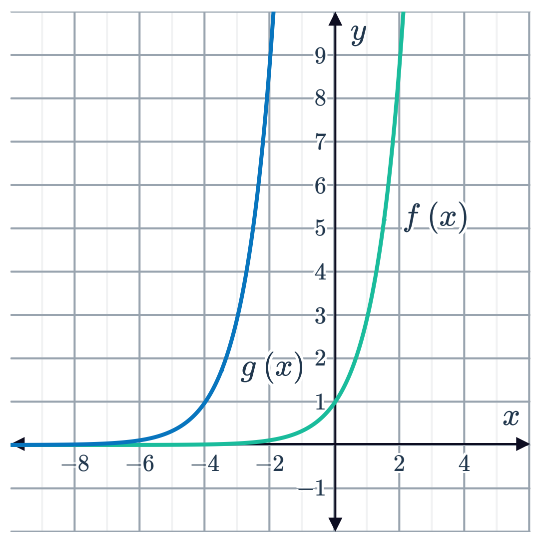
b State the equation of $g(x)$.



9 Consider the graphs of $f(x) = 3^x$ and $g(x)$:

a Write $g(x)$ in terms of $f(x)$.

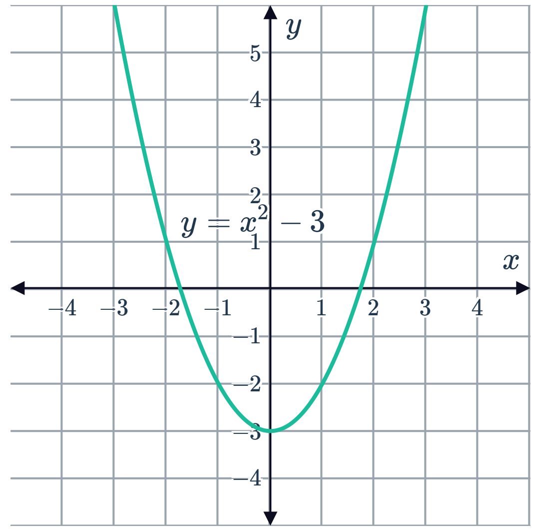
b State the equation of $g(x)$.



10 Consider the graph of $y = x^2 - 3$:

a Write down the new equation when the graph of $y = x^2 - 3$ is moved upwards by 5 units.

b Write down the new equation when the graph of $y = x^2 - 3$ is moved to the right by 3 units.

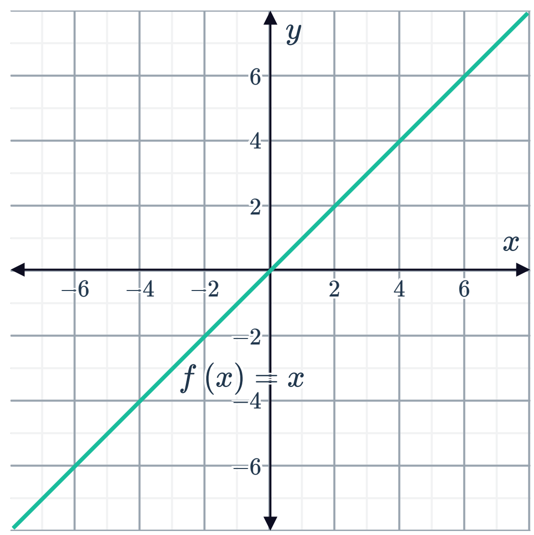


11 Consider the graph of $f(x) = x$.

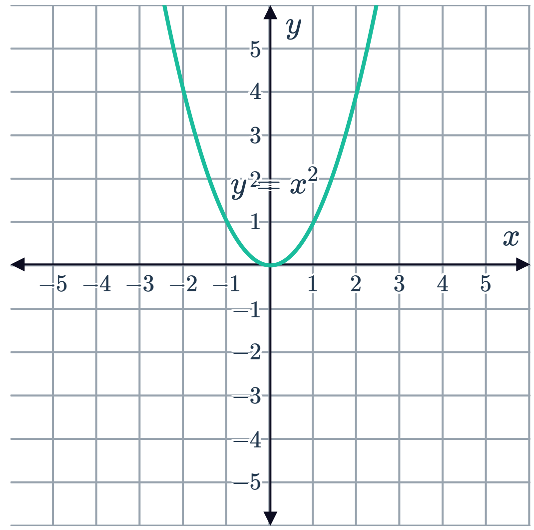
a Write down the equation of the new function $g(x)$ which is formed by evaluating $f(x) + 4$.

b Sketch the graph of $y = g(x)$.

c Describe the transformation the graph of $f(x)$ undergone to become the graph of $g(x)$.



12 Consider the graph of $y = x^2$:



a

i Describe the transformation of the graph of $y = x^2$ that would obtain $y = x^2 - 3$.

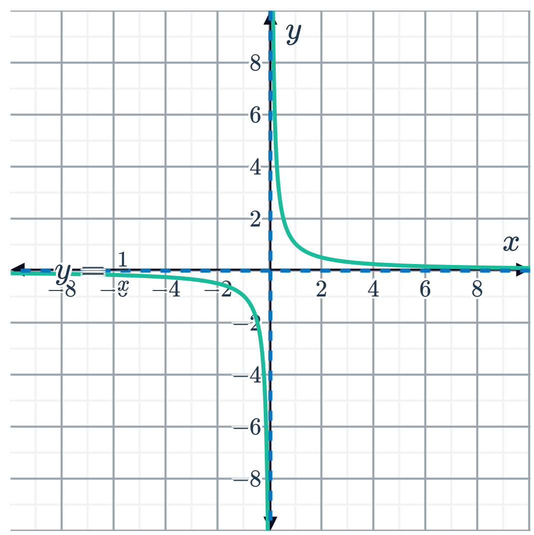
ii Sketch the graph of $y = x^2 - 3$ on the same coordinate plane.

b

i Describe the transformation of the graph of $y = x^2$ that would obtain $y = (x-2)^2$.

ii Sketch the graph of $y = (x-2)^2$ on the same coordinate plane.

13 Consider the graph of $y = \dfrac{1}{x}$:



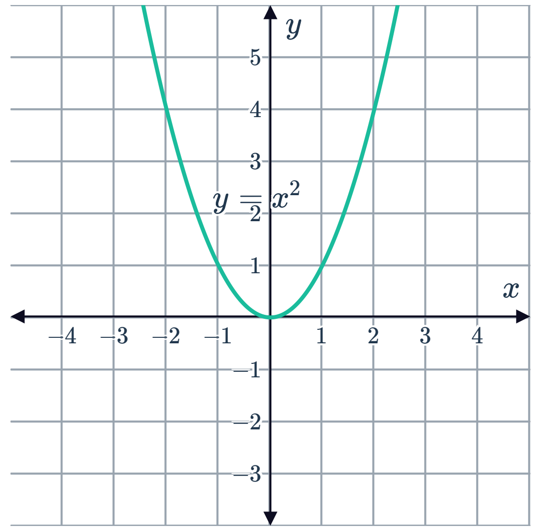
a Describe the transformation of the graph of $y = \dfrac{1}{x}$ that would obtain $y = \dfrac{1}{x+2}$.

b Sketch the graph of $y = \dfrac{1}{x+2}$ on the same coordinate plane.

14 Consider the graph of $y = x^2$:

a Describe the transformation of the graph of $y = x^2$ that would obtain $y = (x-2)^2$.

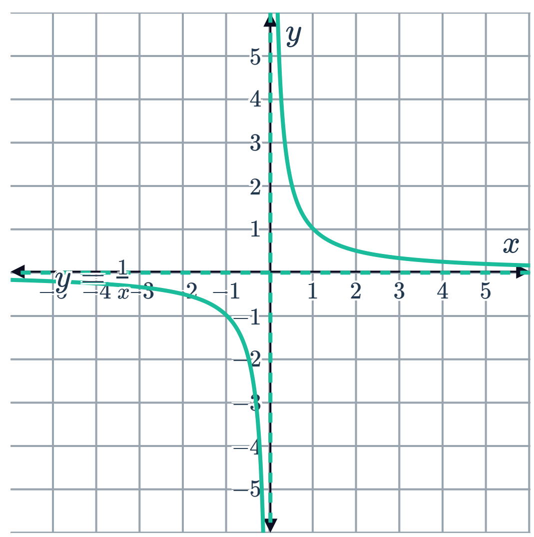
b Sketch the graph of $y = (x-2)^2$.



15 Consider the graph of $y = \dfrac{1}{x}$:

a Describe the transformation of the graph of $y = \dfrac{1}{x}$ that would obtain $y = \dfrac{1}{x} + 3$.

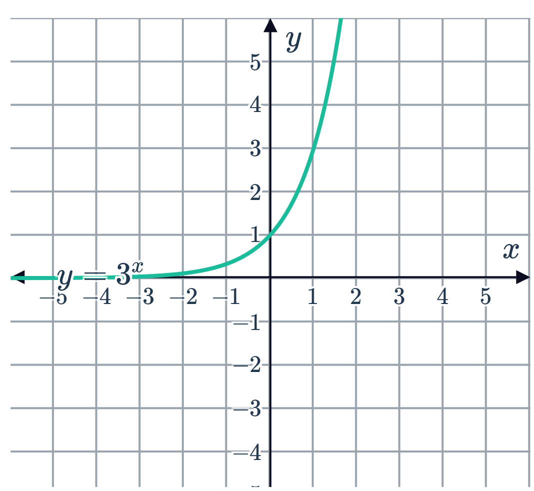
b Sketch the graph of $y = \dfrac{1}{x} + 3$.



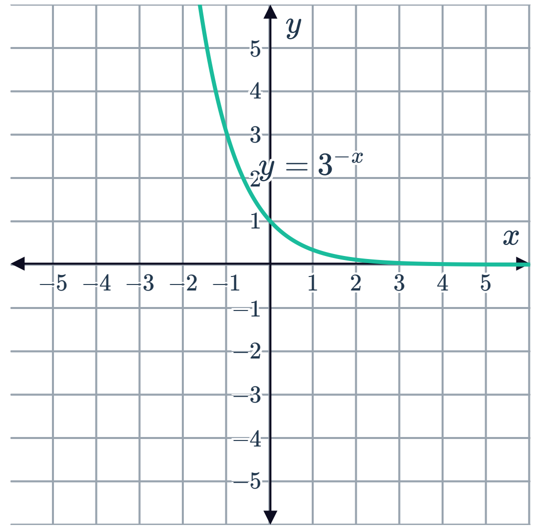
16 Consider the graph of $y = 3^x$:

a Describe the transformation of the graph of $y = 3^x$ that would obtain $y = 3^x - 4$.

b Sketch the graph of $y = 3^x - 4$.



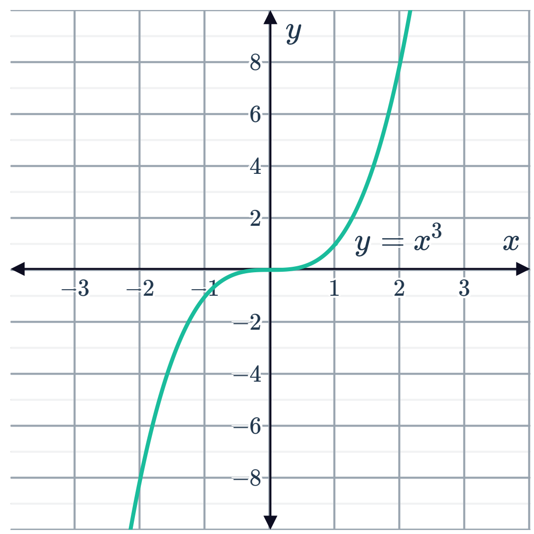
17 Consider the graph of $y = 3^{-x}$:



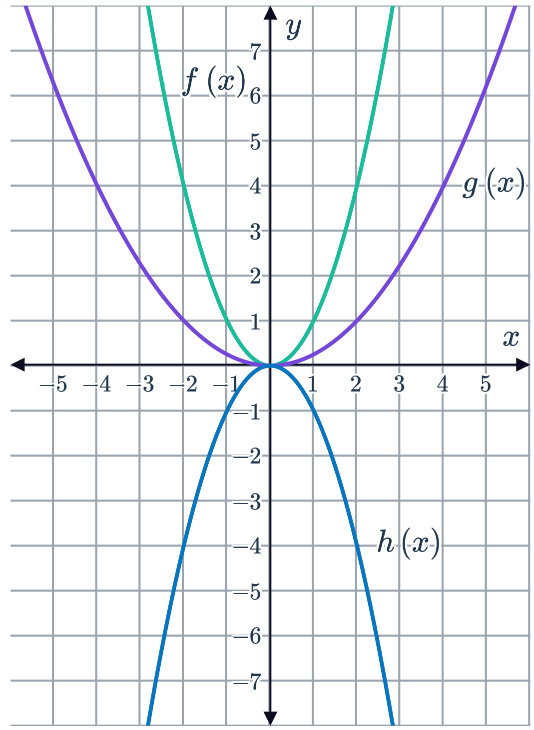
a Describe the transformation of the graph of $y = 3^{-x}$ that would obtain $y = 3^{-x} + 2$.

b Sketch the graph of $y = 3^{-x} + 2$.

18 Consider the graph of $y = x^3$:
Sketch the graph after it has undergone transformations resulting in the function $y = x^3 - 4$.



19 Three functions have been graphed on the coordinate plane. $g(x)$ and $h(x)$ are both transformations of $f(x)$.



a State the equation of $f(x)$.

b Write $g(x)$ in terms of $f(x)$ using the transformation shown in the graph.

c State the equation of $g(x)$.

d Write $h(x)$ in terms of $f(x)$ using the transformation shown in the graph.

e State the equation of $h(x)$.

Multiple transformations

20 Describe the transformations that have occured from $f(x)$ to $g(x)$:

a From $f(x) = x^4$ to $g(x) = -5x^4 + 8$

b From $f(x) = x^4$ to $g(x) = -7(x+4)^4$

c From $f(x) = x^2$ to $g(x) = -10(x+8)^2 - 9$

21 The graph of $y = -x^2$ is moved to the right by 6 units and upwards by 5 units. What is the new equation of the graph?

22 Suppose that $(-4, 3)$ is a point on the graph of $y = g(x)$. Find the corresponding point on the graph of:

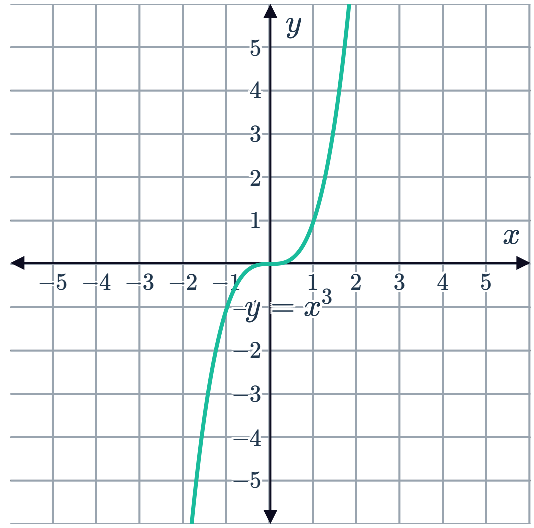
a $y = g(x+7) - 6$

b $y = -6g(x-4) + 6$

c $y = g(6x+1)$

23 Consider the function $f(x) = \sqrt{x}$. Write down the new function $g(x)$ which results from compressing $f(x)$ vertically by a factor of $\dfrac{1}{3}$, and compressing horizontally by a factor of $\dfrac{1}{2}$.

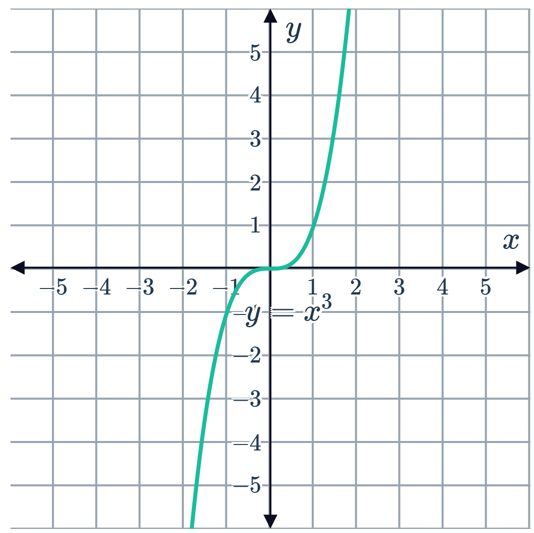
24 Consider the graph of $y = x^3$:



a Describe the transformation of the graph of $y = x^3$ that would obtain $y = (x+2)^3 + 4$.

b Sketch the graph of $y = (x+2)^3 + 4$.

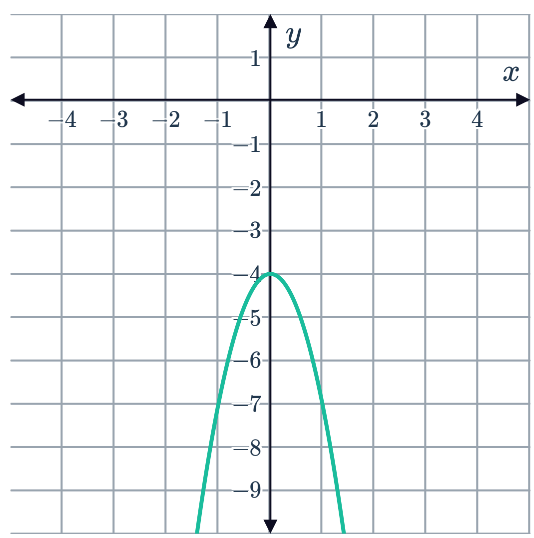
25 Consider the graph of $y = x^3$:



a Describe the transformation of the graph of $y = x^3$ that would obtain $y = (x-2)^3 - 3$.

b Sketch the graph of $y = (x-2)^3 - 3$.

26 Consider the following graph:



a Describe the transformations of the graph of $y = x^2$ to the given graph.

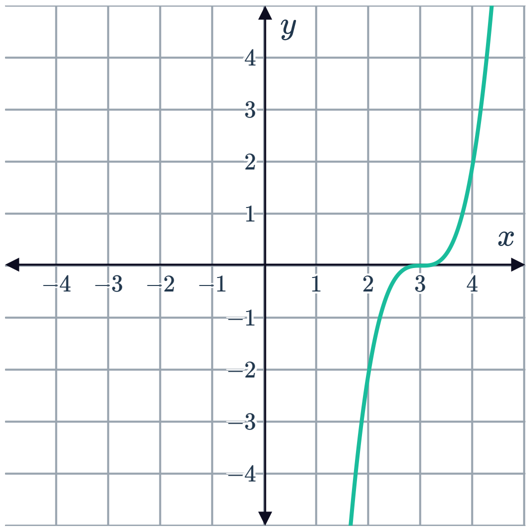
b Write down the equation of the given graph.

27 For each of the following graphs:

i Describe the transformations of the graph of $y = x^3$ to the given graph.

ii Write down the equation of the given graph.

a



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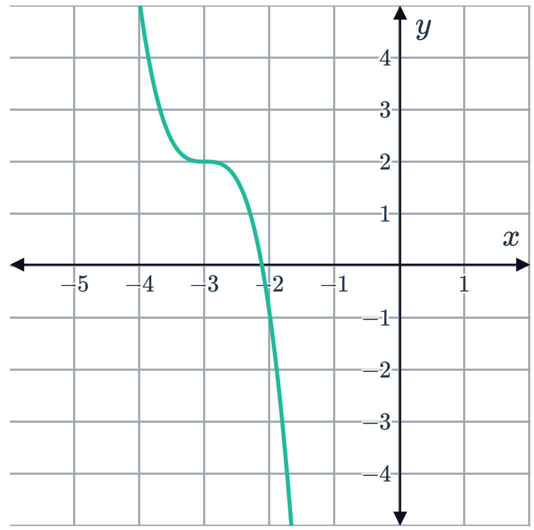
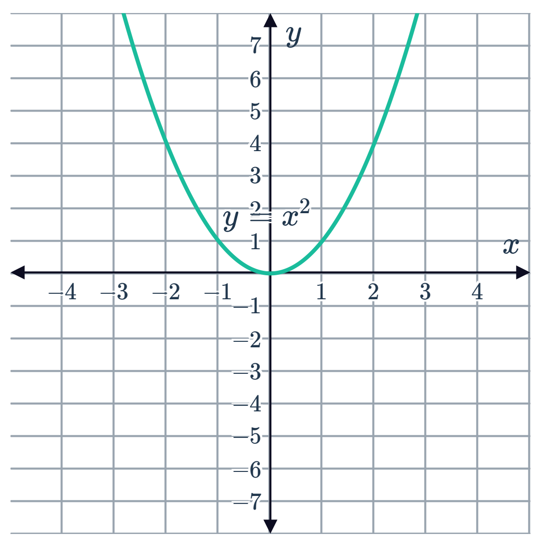


28 Consider the graph of $y = x^2$:
Sketch the graph after it has undergone transformations resulting in the function $y = 4x^2 - 2$.



29 Consider the function $f(x) = \sqrt{x}$.

a Complete the table for $f(x)$:

b Sketch a graph of the function $f(x)$.

x	0	1	4	9	16
$f(x)$					
$g(x)$					

c Complete the table of values for the transformed function $g(x) = f(4x)$, giving all values in exact form.

d Sketch a graph of $g(x)$.

e Describe how $g(x)$ relates to the graph of $f(x)$.

f Explain why the transformed function $h(x) = f(-4x)$ can be undefined for the values of x in the table.

30 The table below shows values that satisfy the function $f(x) = |x|$:

x	-3	-2	-1	0	1	2	3
$y = f(x)$	3	2	1	0	1	2	3

a Complete the table of values for each transformation of the function $f(x)$:

x	-3	-2	-1	0	1	2	3
$y = f(x)$	3	2	1	0	1	2	3
$g(x) = 5f(x)$							
$h(x) = -2f(x)$							

b Sketch the graphs of $f(x)$ and $g(x)$ on the same coordinate plane.

c Describe the transformation undergone by $f(x)$ to become $g(x)$.

d Sketch the graphs of $f(x)$ and $h(x)$ on the same coordinate plane.

e Describe the transformation undergone by $f(x)$ to become $h(x)$.